

Program: Diploma in Food Processing Technology	
Course Code: 4427	Course Title: Food Microbiology II Lab
Semester: 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:1)	Periods per semester: 45

Course Objectives:

- To evaluate the microbiological quality of samples
- To perform qualitative and quantitative microbial examination of samples
- To identify various microorganisms from the sample.
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Course Prerequisites:

Topic	Course Code	Course Name	Semester
Awareness about microorganism		Microbiology lab I	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Evaluate the total microbial load in the food or in the environment	14	Applying
CO2	Identify suitable methods for qualitative and quantitative evaluation of food	13	Applying
CO3	Demonstrate the procedure for identification of specific microorganism in food	11	Applying
CO4	Demonstrate the usage of beneficial microorganism for production of food	7	Applying

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3	3		
CO2				3	3		
CO3				3	3		
CO4				3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Evaluate the total microbial load in the food or in the environment		
M1.01	Evaluation of microbiological quality of potable water	2	Understanding
M1.02	Demonstrate Presumptive test for Coliforms in milk	3	Applying
M1.03	Evaluate of microbiological quality of milk (MBRT)	3	Applying
M1.04	Evaluate of microbiological quality of food products	3	Applying
M1.05	Estimate of microbial count of air	3	
CO2	Identify suitable methods for qualitative and quantitative evaluation of food		
M2.01	Demonstrate Microbial Examination of Fruit sample- surface washing and internal tissue	3	Applying
M2.02	Demonstrate Microbial Examination of Vegetable sample- surface washing and internal tissue	3	Applying
M2.03	Quantitative analyze the milk by Standard plate count (SPC) method.	2	Applying
M2.04	Perform Counting for yeasts and moulds	3	Applying
	Lab Test –I	2	
CO3	Demonstrate the procedure for identification of specific microorganism in food		

M3.01	Methods for identification of Escherichia coli	2	Applying
M3.02	Methods for identification of Staphylococcus aureus	3	Applying
M3.03	Methods for identification of Bacillus cereus	3	
M3.04	Methods for identification of Vibrio cholerae	3	
CO4	Demonstrate the usage of beneficial microorganism for production of food		
M4.01	Demonstration of microbial production of curd	2	Applying
M4.02	Demonstration of microbial production of wine	3	Applying
	Lab Test– II	2	

Text/Reference

T/R	BookTitle/Author
R1	James Cappuccino, Microbiology: A Laboratory Manual, 10th edition., Natalie Sherman. Pearson Higher education
R2	R C Dubey and D K Maheshwary, Practical Microbiology, S. Chand & Co. Ltd., 2006
R3	Prescott's Microbiology w/ Harley Lab Manual 8th Ed. Joanne Wiley. McGraw Hill education
R4	Bibek Ray, Fundamental Food Microbiology, CRC press

Online Resources

Sl.No	WebsiteLink
1.	https://vlab.amrita.edu/?sub=3&brch=73&sim=1630&cnt=1
2.	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4846335/
3.	https://sites.law.duke.edu/forensicsforum/2017/03/17/an-overview-of-presumptive-and-confirmatory-testing-in-the-forensic-sciences/